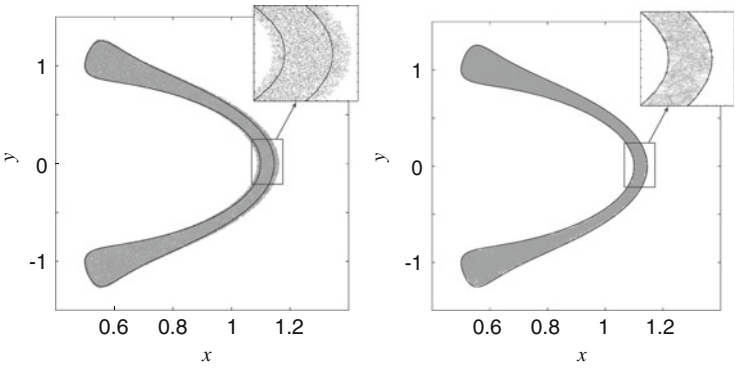


Fig. 5.8 Long trajectories of the isosceles 3-body problem computed using two different timesteps, $h = 0.07$ (*left*) and $h = 0.01$ (*right*). The discrete values have been used to draw *small dots* which appear to fill in a region bounded by a level curve of the potential energy, with some error at the boundary that decreases with stepsize h

have plotted two long numerical solutions (100,000 timesteps) in their xy projection for two different stepsizes $h = 0.07$ and $h = 0.01$, started from a point with energy $\mathrm{E} = -0.5$ which is (a) admissible and (b) connected and (c) bounded. The dark curve represents the level curve of the potential energy for $U = -0.5$ while the numerical steps are shown in gray. As we can see in this figure, the points appear to shade the region completely, however spilling over a bit from the interior of the curve. This overspill is due to the numerical error, as it is decreased by a decrease in the stepsize.

As we have seen in the Chap. 3, when a symplectic method is applied to a molecular dynamics problem it induces a perturbed Hamiltonian (energy) function. For the Verlet scheme the modified Hamiltonian is

$$\tilde{H}_h = H + \frac{h^2}{24}\left(2\boldsymbol{p}^T\boldsymbol{M}^{-1}U''(\boldsymbol{q})\boldsymbol{M}^{-1}\boldsymbol{p} - \nabla U(\boldsymbol{q})\boldsymbol{M}^{-1}\nabla U(\boldsymbol{q})\right) + \mathcal{O}(h^4).$$

Actually, as explained previously, we must interpret this carefully: a finite truncation of this expansion may characterize the discrete trajectories observed in practice, but the series expansion itself does not typically converge. Assuming such a perturbative expansion is valid, the issue of ergodicity of the numerical method can be related to the corresponding issue for the dynamical system characterized by $\tilde{H}_h$.

A result in [338] provides a formula to calculate the perturbation of the microcanonical average induced by the Hamiltonian perturbation. This result has been recast in the more convenient form in [14]:

Theorem 5.1 *Let $H(\boldsymbol{q},\boldsymbol{p})$ and $\tilde{H}(\boldsymbol{q},\boldsymbol{p}) = H(\boldsymbol{q},\boldsymbol{p}) + \varepsilon\eta(\boldsymbol{q},\boldsymbol{p})$ be two Hamiltonian functions. Suppose the energy surface $\Sigma_{\mathrm{E}} = \{(\boldsymbol{q},\boldsymbol{p})|H(\boldsymbol{q},\boldsymbol{p}) = \mathrm{E}\}$ to be compact and define the perturbed energy surface by $\tilde{\Sigma}_{\mathrm{E}} = \{(\boldsymbol{q},\boldsymbol{p})|\tilde{H}(\boldsymbol{q},\boldsymbol{p}) = \mathrm{E}\}$, also assumed compact.*